GOOGLE CLOUD SKILLS | HANDS-ON LAB REPORT

Google Cloud Storage Bucket Lock

Retention policies and holds · Lab ID: GSP297

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DATE COMPLETED	15 October 2026
PLATFORM	Google Cloud (Cloud Storage, Cloud Shell)
LEVEL / DURATION	Introductory / 10 minutes
FINAL SCORE	100 / 100

1. Overview

The Cloud Storage **Bucket Lock** feature lets you set a **data retention policy** on a bucket that controls how long objects must be kept. The policy can then be **locked**, which permanently prevents it from being reduced or removed. Combined with detailed audit logging or Object Lifecycle Management, Bucket Lock supports regulatory and compliance needs such as FINRA, SEC and CFTC rules and healthcare retention requirements.

The lab follows a branch of a financial institution that must retain transaction records under SEC Rule 17a-4. The 6-year requirement is shortened to 10 seconds for the lab. All work was done with `gcloud storage` commands in Cloud Shell.

2. Objectives

- Create a bucket
- Define and lock an object retention policy
- Use temporary holds and event-based holds
- Delete a bucket that has a locked retention policy

3. Concepts and Resources

Concept	What it does
Retention policy	Sets a minimum time every object must be kept before it can be deleted or overwritten. Here: 10 seconds.
Locked policy	Once locked, the policy can be extended but never reduced or removed.
Temporary hold	A flag on an object that blocks deletion until the hold is released, regardless of the object's age.
Event-based hold	Holds an object and keeps its retention clock from starting until the hold is released (for example, when a loan is paid off).

Resource	Details
Project / bucket	qwiklabs-gcp-00-89a092e9a52e (the bucket is named after the project ID)
Retention period	10 seconds, then locked
dummy_transactions	45 bytes, text/plain, used to test the retention policy and the temporary hold
dummy_loan	28 bytes, application/octet-stream, used to test the event-based hold

The bucket was deleted at the end of the lab.

4. Step-by-Step Walkthrough

Task 1 – Create a bucket

I stored the project ID in an environment variable and created a bucket named after it:

```
export BUCKET=$(gcloud config get-value project)
gcloud storage buckets create "gs://$BUCKET"
```

Task 2 – Define a retention policy

I set a 10-second retention policy, checked it, copied a sample transaction record into the bucket, and listed the object to see its retention expiration:

```
gcloud storage buckets update "gs://$BUCKET" --retention-period=10s
gcloud storage buckets describe "gs://$BUCKET" --format="default(retention_policy)"
gcloud storage cp gs://spls/gsp297/dummy_transactions "gs://$BUCKET/"
gcloud storage ls -L "gs://$BUCKET/dummy_transactions"
```

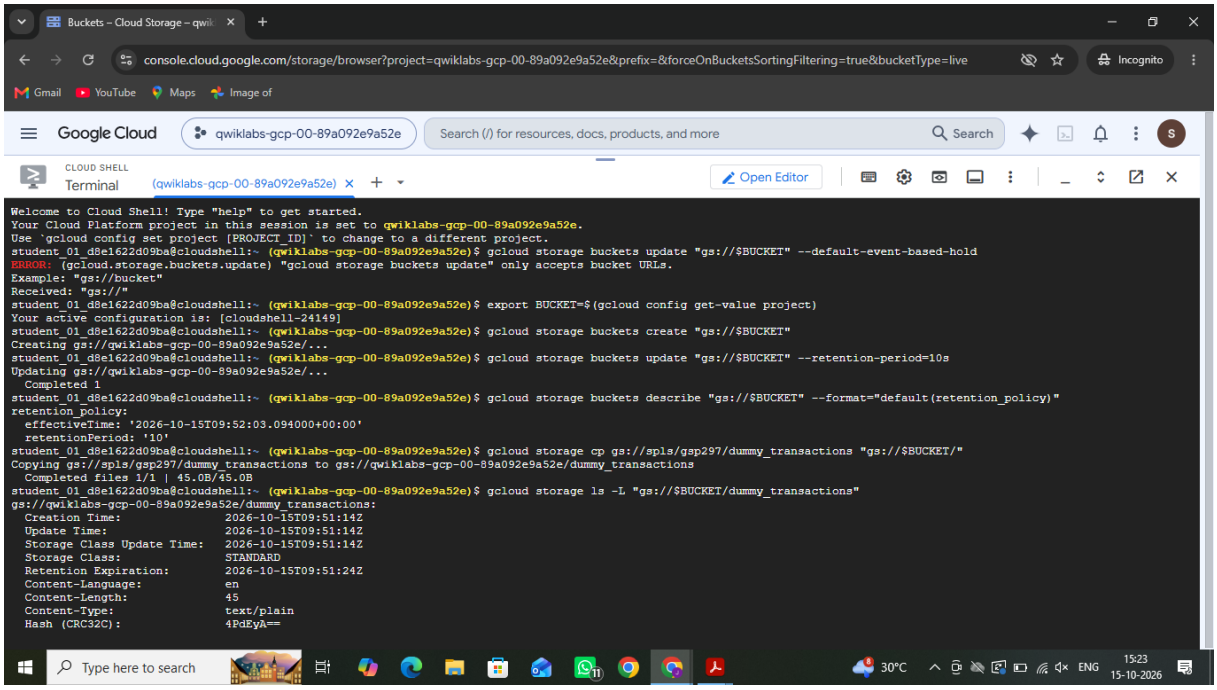


Figure 1: Cloud Shell – bucket created, retention period set to 10s and verified (retentionPeriod: '10'), dummy_transactions copied, and its listing showing Retention Expiration 2026-10-15T09:51:24Z, ten seconds after creation. The first command at the top failed because it was typed before $BUCKET was defined.

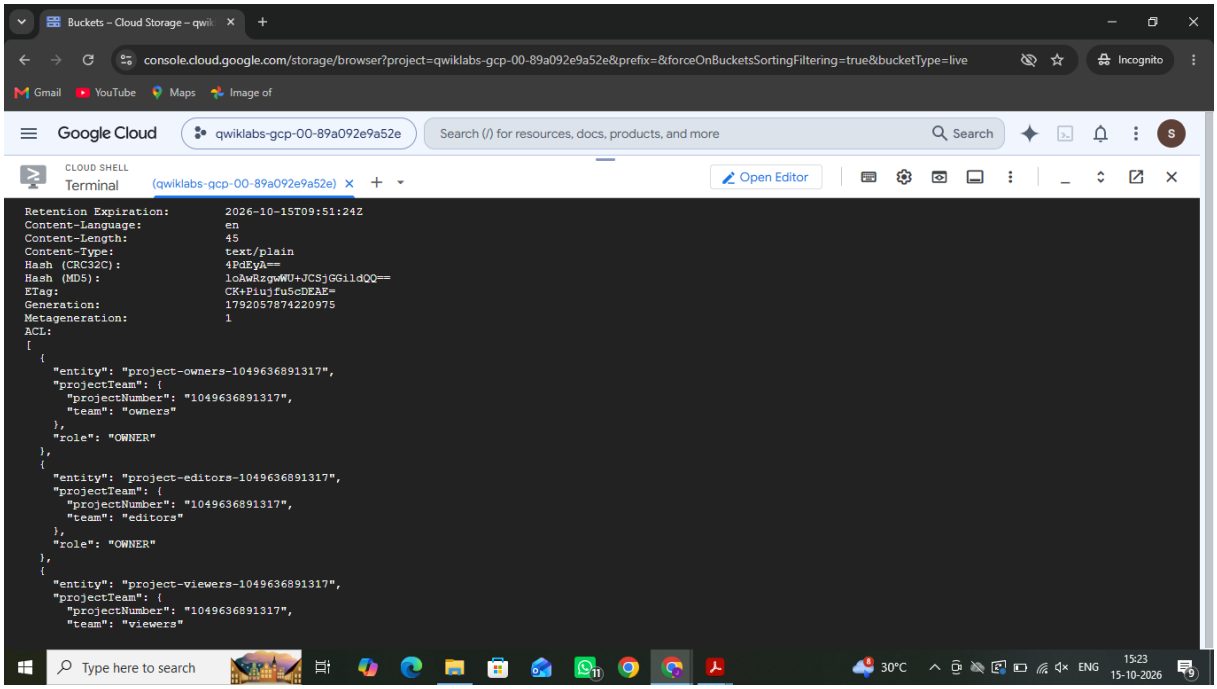


Figure 2: Rest of the object listing – 45-byte text/plain object, generation, metageneration 1 and its ACL entries.

Task 3 – Lock the retention policy

```
gcloud storage buckets update "gs://$BUCKET" --lock-retention-period
```

The command warns that the setting is permanent and asks for confirmation. My first attempt was cancelled with Ctrl+C at the prompt, so I ran it again and answered **y**. The output ends with "Locking retention policy" and "Completed 1".

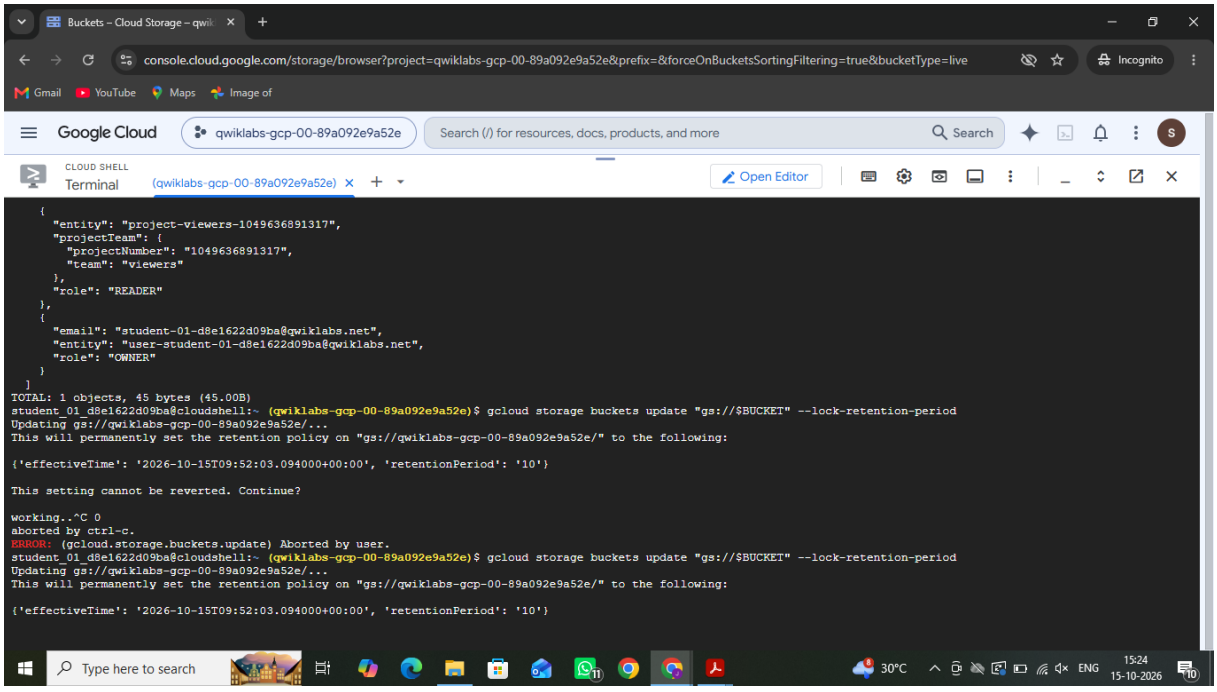


Figure 3: Lock command and its warning ("This setting cannot be reverted"). The first run was aborted at the prompt; the second run is shown being repeated.

Task 4 – Temporary hold

I placed a temporary hold on the transaction record and tried to delete it. Cloud Storage refused with an HTTP 403 error, because the object is under an active temporary hold. After releasing the hold, the delete succeeded.

```
gcloud storage objects update "gs://$BUCKET/dummy_transactions" --temporary-hold
gcloud storage rm "gs://$BUCKET/dummy_transactions"           # 403: active Temporary hold
gcloud storage objects update "gs://$BUCKET/dummy_transactions" --no-temporary-hold
gcloud storage rm "gs://$BUCKET/dummy_transactions"           # succeeds
```

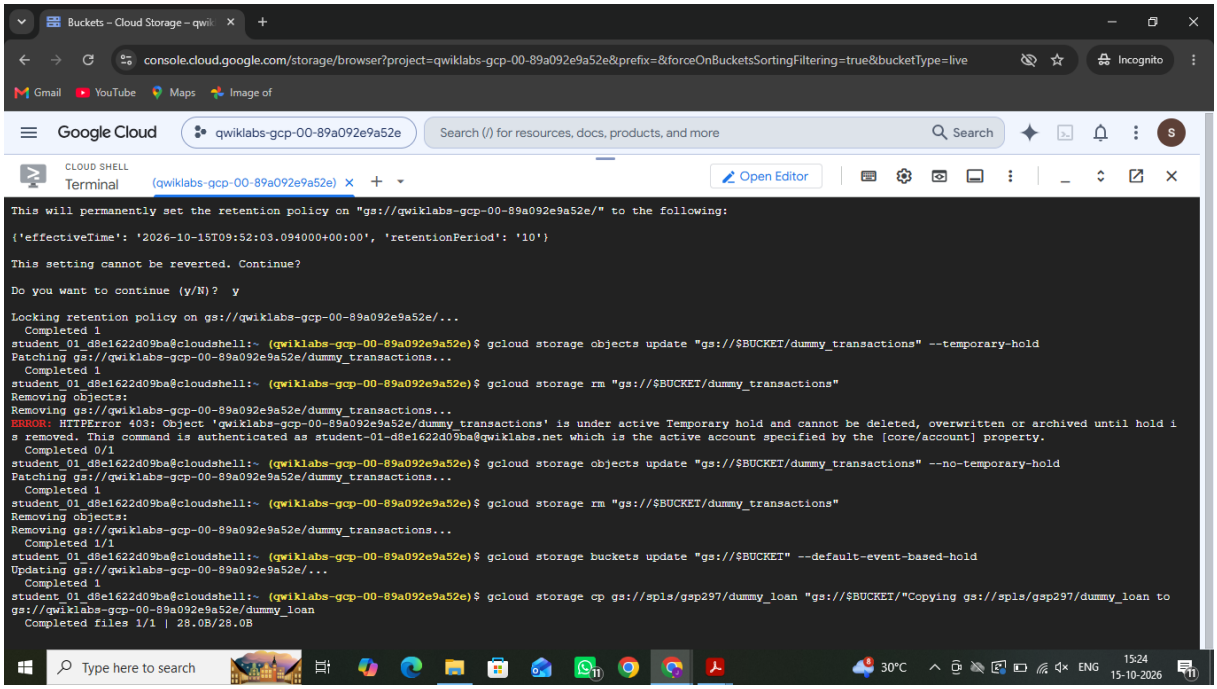


Figure 4: Policy locked (answered y), temporary hold set, delete blocked with "HTTPError 403 ... under active Temporary hold", hold released, delete completed (1/1). The same screen shows the default event-based hold enabled and dummy_loan copied in.

Task 5 – Event-based hold

Next I enabled a default event-based hold on the bucket, so new objects start out held, and added a sample loan record:

```
gcloud storage buckets update "gs://$BUCKET" --default-event-based-hold
gcloud storage cp gs://spls/gsp297/dummy_loan "gs://$BUCKET/"
gcloud storage ls -L "gs://$BUCKET/dummy_loan"
```

While the event-based hold is on, the object has no retention expiration. When the loan is paid off, the hold is released and the bucket's 10-second retention policy starts counting:

```
gcloud storage objects update "gs://$BUCKET/dummy_loan" --no-event-based-hold
gcloud storage ls -L "gs://$BUCKET/dummy_loan"
```

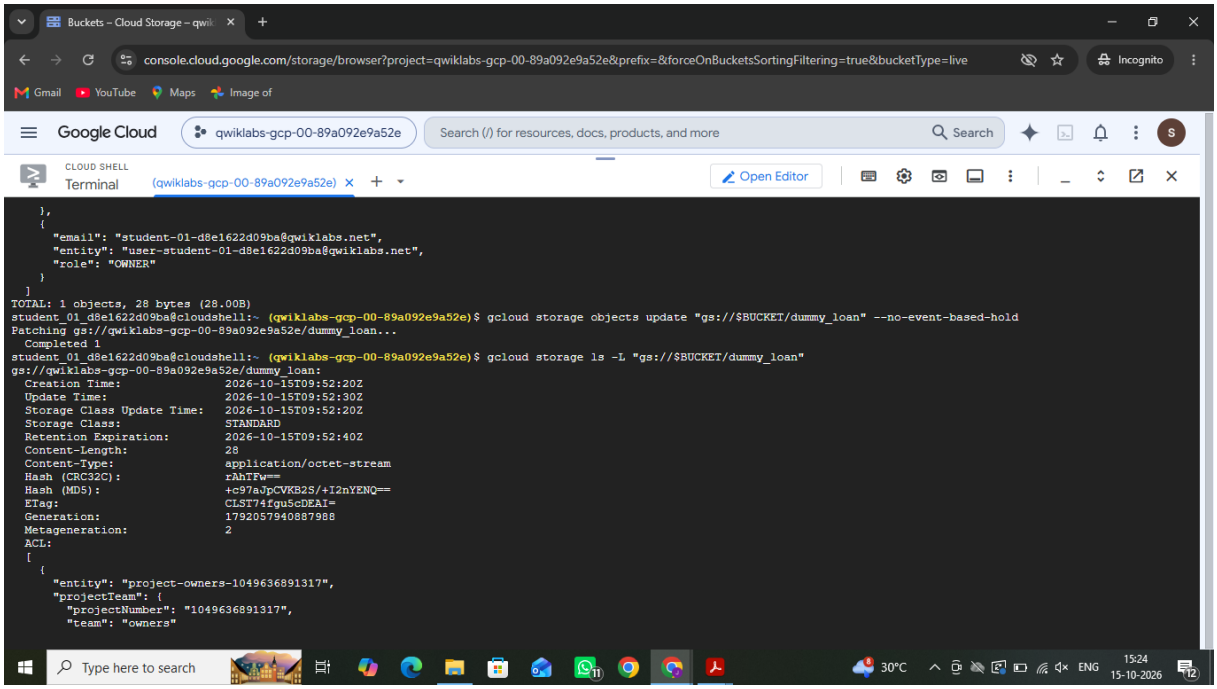


Figure 5: Event-based hold released on dummy_loan. The listing now shows a Retention Expiration of 2026-10-15T09:52:40Z, 10 seconds after the hold was released (update time 09:52:30Z), and metageneration 2 after the update.

Task 6 – Delete the bucket

Once the loan record's retention period had expired, I deleted the object and then the bucket. A bucket with a locked retention policy can still be deleted when it is empty, because it holds no data that is subject to retention.

```
gcloud storage rm "gs://$BUCKET/dummy_loan"
gcloud storage buckets delete "gs://$BUCKET/"
```

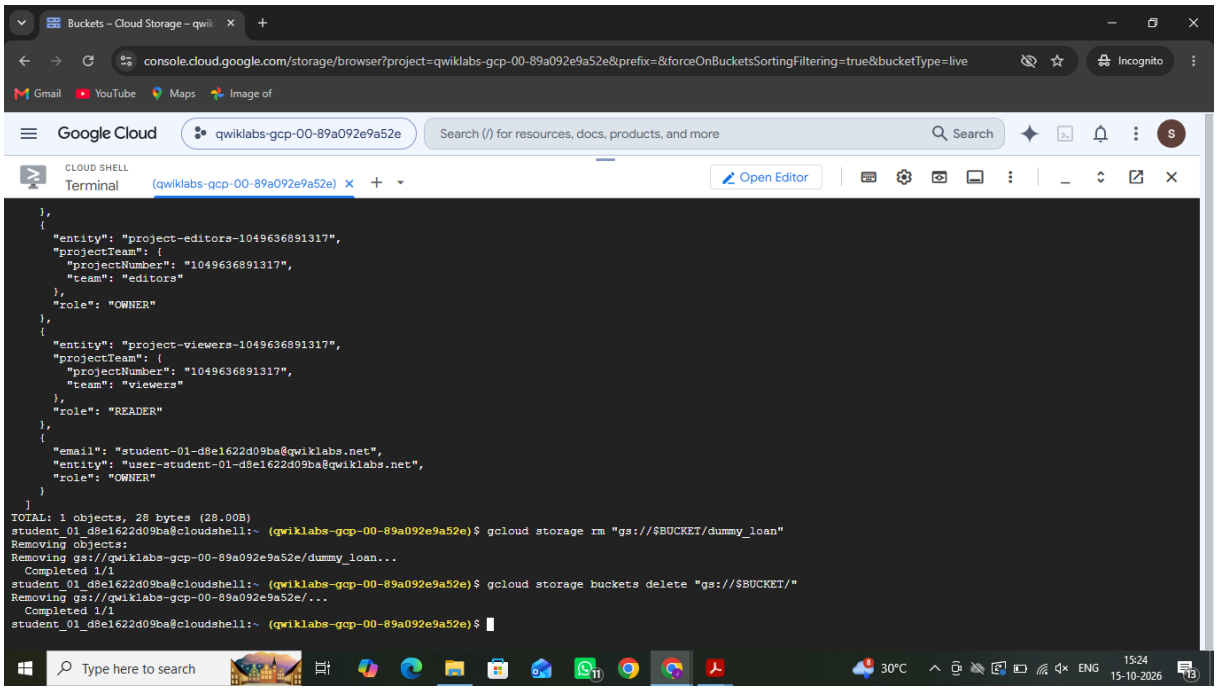


Figure 6: dummy_loan removed (Completed 1/1) and the empty bucket deleted (Completed 1/1).

5. Lab Completion

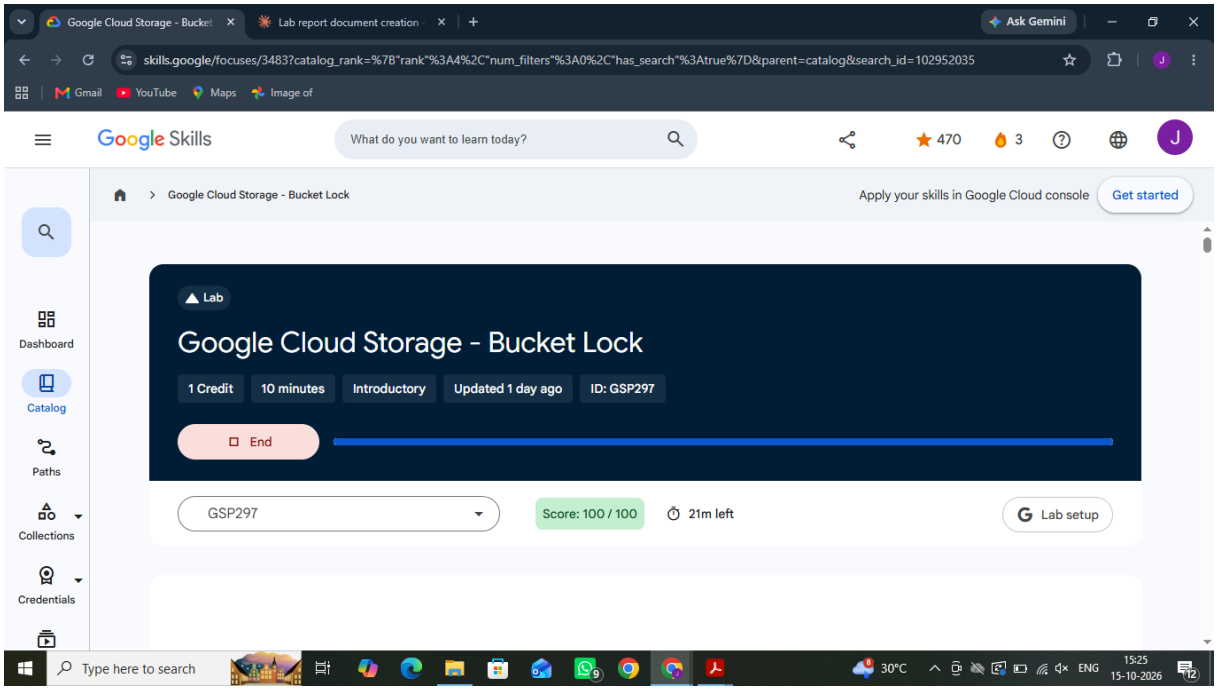


Figure 7: Lab completed with a score of 100/100.

6. Key Takeaways

- A **retention policy** sets the minimum time objects must be kept. It can be defined in seconds, days, months or years.
- **Locking** a retention policy is permanent. It can only be extended, never reduced or removed, which is what makes Bucket Lock useful for compliance.
- A **temporary hold** blocks deletion of an object, even after its retention period, until the hold is released. A delete attempt returns HTTP 403.
- An **event-based hold** keeps the retention clock from starting. When the hold is released (for example, when a loan is repaid), the retention period starts from that moment.
- A bucket with a locked retention policy can be deleted only when it is empty, so objects must reach the end of their retention period and holds first.
- Always read the confirmation prompt on irreversible commands such as `--lock-retention-period` before answering y.

7. Conclusion

The lab was completed with a score of 100/100. I created a bucket, applied and locked a 10-second retention policy, tested temporary and event-based holds, and removed the bucket, which covers the main controls Cloud Storage offers for regulatory data retention.